

DATA VERACITY: GENERATIVE AI 2

Advanced Cases in Assurance Services (ACCTG 521)
Class 14 | MPAcc class of 2026

Agenda

Review: Individual Deliverable (Due Nov 13th)

Labs:

- ETL for Multiple Input Files
- Preliminary Assessments of Gen AI Sophistiation
- Time for Memo Writeup

Swindler Case Part 2



FOSTER
Master of
Professional Accounting

Part Two Requirements

Perform at least seven tests that will help determine if the data set is real or fake. Your tests can use the data included in the file Skylar sent you or you can add external data. If you add external data, include the link in the test explanation showing where you acquired the data.

Memo Requirements

- Documentation of Preliminary Testing Environment (last class)
- Outline of Expectations (more sophisticated Generative AI, last or this class)
- Analysis of Increasing Gen AI sophistication and implications (this class)

Memorandum¶

To: Asher-Curtis¶

From: {Your-Name}, Data-Veracity-Specialist¶

Topic: Building-a-Testing-Environment-for-Generative-AI-Fabricated-Data¶
¶

Preliminary-Analysis¶

Two-Paragraphs-to-half-a-page-on-the-preliminary-analyses-run-and-key-takeaways.·

Reference-Appendix-1, "Top-7-Preliminary-Tests-on-Pure-Oils-Test-Data"¶

¶

Expectations-for-Subsequent-Analysis¶

One-to-two-paragraphs-on-your-expectations-about-the-subsequent-analysis-including-(but-not-limited-to)-potential-automation-of-testing-any-number-of-files-using-the-logic-from-the-preliminary-analyses-run; expectations-about-how-changes-in-Gen-AI-sophistication-can-impact-input-data-and-testing; any-other-trade-offs-or-expected-challenges-including-ETL.¶

¶

Gen-AI-Sophistication-Analysis¶

Up-to-half-a-page-of-discussion-of-the-Gen-AI-sophistication-analyses-and-key-take-aways.·

Discuss-the-trends/features/takeaways-of-increasing-Gen-AI-sophistication-in-the-input-data-had-on-the-testing-environment. Reference-Appendix-2.¶

¶

Conclusions¶

Conclude-your-analysis.¶

Directly from Case

+ Ask the Right Questions

Appendix 1:¶

Top 7 Preliminary Tests on Pure Oils Test Data¶

Test #¶	Expectation (include why)¶	Test (with references)¶	Conclusion¶	Questions to Ask¶
1¶	¶	¶	¶	¶
2¶	¶	¶	¶	¶
3¶	¶	¶	¶	¶
4¶	¶	¶	¶	¶
5¶	¶	¶	¶	¶
6¶	¶	¶	¶	¶
7¶	¶	¶	¶	¶

¶

Page Break ¶

Based on the conclusion from the test, what question would you ask the hypothetical manager / borrower?

Memorandum¶

To: Asher-Curtis¶

From: {Your-Name}, Data-Veracity-Specialist¶

Topic: Building-a-Testing-Environment-for-Generative-AI-Fabricated-Data¶

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¶

Conclusions¶

Conclude-your-analysis.¶

If you didn't complete this, change from expectations to actual ETL challenges, if you did: footnote actual ETL challenges

Appendix 2

Depending on how you have set up your analysis:

- Tabulate the results for submission types 1, 2, & 3
- OR
- Take screenshots of the test outcomes you want to discuss.
- E.g., The more sophisticated prompt 2, did not fail test X as often as prompt 1, see Appendix 2 Table/Figure X
- Write up as many of the tests as you feel support your conclusion, but don't ignore contrary evidence.

In the Memo Text for Gen AI section

Link your discussion to the Appendix:

- E.g., The more sophisticated prompt 2, did not fail test X as often as prompt 1, see Appendix 2 Table/Figure X
- Write up as many of the tests as you feel support your conclusion, but don't ignore contrary evidence.

Finally

Please upload your workflow(s) used in these two classes to the same portal as the memo

<https://canvas.uw.edu/courses/1826714/assignments/10826325>

Agenda

Review: Individual Deliverable (Due Nov 13th)

Labs:

- **ETL for Multiple Input Files**
- Preliminary Assessments of Gen AI Sophistiation
- Time for Memo Writeup

ETL for multiple files

ETL Steps:

1. ETL selected data
2. Update your file to accommodate multiple files (or run them one-by-one)
3. Run your analysis for each level of sophistication.

****Note:** In addition to your own file(s) you can git pull the class submissions to ACCTG 522.

submission_1_1.csv	submission_1_15.csv	submission_1_29.csv	submission_2_14.csv	submission_3_1.csv
submission_1_2.csv	submission_1_16.csv	submission_2_1.csv	submission_2_15.csv	submission_3_2.csv
submission_1_3.csv	submission_1_17.csv	submission_2_2.csv	submission_2_16.csv	submission_3_3.csv
submission_1_4.csv	submission_1_18.csv	submission_2_3.csv	submission_2_17.csv	submission_3_4.csv
submission_1_5.csv	submission_1_19.csv	submission_2_4.csv	submission_2_18.csv	submission_3_5.csv
submission_1_6.csv	submission_1_20.csv	submission_2_5.csv	submission_2_19.csv	submission_3_6.csv
submission_1_7.csv	submission_1_21.csv	submission_2_6.csv	submission_2_20.csv	
submission_1_8.csv	submission_1_22.csv	submission_2_7.csv	submission_2_21.csv	
submission_1_9.csv	submission_1_23.csv	submission_2_8.csv	submission_2_22.csv	
submission_1_10.csv	submission_1_24.csv	submission_2_9.csv	submission_2_23.csv	
submission_1_11.csv	submission_1_25.csv	submission_2_10.csv	submission_2_24.csv	
submission_1_12.csv	submission_1_26.csv	submission_2_11.csv	submission_2_25.csv	
submission_1_13.csv	submission_1_27.csv	submission_2_12.csv	submission_2_26.csv	
submission_1_14.csv	submission_1_28.csv	submission_2_13.csv	submission_2_27.csv	

**64 potential
input files**

Record	UniqueID	LastName	FirstName	JobTitle	Department	HireDate	LastPromotionDate	TerminationDate	AnnualSalary	FileName
1	1	Swindler	Skylar	CEO	Executive	2021-01-01	[Null]	[Null]	250000.0	submission_1_1
2	2	Baxter	Colton	HR Coordinator	HR	2021-04-29	2023-10-27	[Null]	178602.88	submission_1_1
3	3	Lynch	Kimberly	Marketing Director	Marketing	2024-02-22	2025-03-20	[Null]	153715.41	submission_1_1
4	4	Underwood	Ryan	Software Engineer	IT	2023-01-15	2024-06-30	[Null]	154360.26	submission_1_1
5	5	Compton	Chad	Marketing Director	Marketing	2023-05-10	2023-10-03	[Null]	61660.68	submission_1_1
6	6	Brown	Margaret	Quality Assurance Tech	Production	2021-01-08	2022-07-24	2023-11-21	113044.71	submission_1_1
7	7	Martinez	William	Production Manager	Production	2025-08-31	2025-09-27	[Null]	63176.43	submission_1_1
8	8	Sanchez	Nathan	Production Manager	Production	2023-04-02	2024-01-09	[Null]	130428.16	submission_1_1
9	9	Bennett	Jessica	Supply Chain Analyst	Operations	2023-05-13	2024-07-30	[Null]	74181.37	submission_1_1
10	10	Contreras	Dustin	Account Executive	Sales	2021-05-02	2025-02-14	2025-05-06	71006.97	submission_1_1
228	EMP008	Gonzalez	Ethan	Sales Associate	Sales	11/23/2018	3/5/2020	[Null]	109429	submission_1_11
229	EMP009	Brown	William	Logistics Coordinator	Operations	5/8/2020	4/26/2023	[Null]	102469	submission_1_11
230	EMP010	Rodriguez	Mia	Training Specialist	HR	9/29/2020	[Null]	[Null]	91500	submission_1_11
231	EMP011	Martin	Mia	CFO	Executive	5/12/2018	6/25/2020	[Null]	235307	submission_1_11
232	EMP012	Garcia	Mason	Account Executive	Sales	8/7/2023	5/26/2026	[Null]	84718	submission_1_11
233	EMP013	Garcia	Harper	SEO Specialist	Marketing	7/22/2024	3/11/2027	[Null]	69373	submission_1_11
234	EMP014	Lee	Emma	Customer Support Rep	Customer Service	7/12/2018	12/15/2019	[Null]	67666	submission_1_11
235	EMP015	Jones	Alexander	Systems Administrator	IT	9/6/2024	3/1/2027	[Null]	71504	submission_1_11
236	EMP016	Smith	Isabella	SEO Specialist	Marketing	10/25/2021	[Null]	[Null]	84735	submission_1_11
237	EMP017	Lee	Sophia	Training Specialist	HR	3/12/2019	1/10/2021	[Null]	88270	submission_1_11
238	EMP018	Lopez	Henry	Recruiter	HR	10/24/2019	4/19/2022	[Null]	94252	submission_1_11

Date formats in the files:

- yyyy-mm-dd
- mm/dd/yyyy
- dd-mm-yyyy etc.

ETL

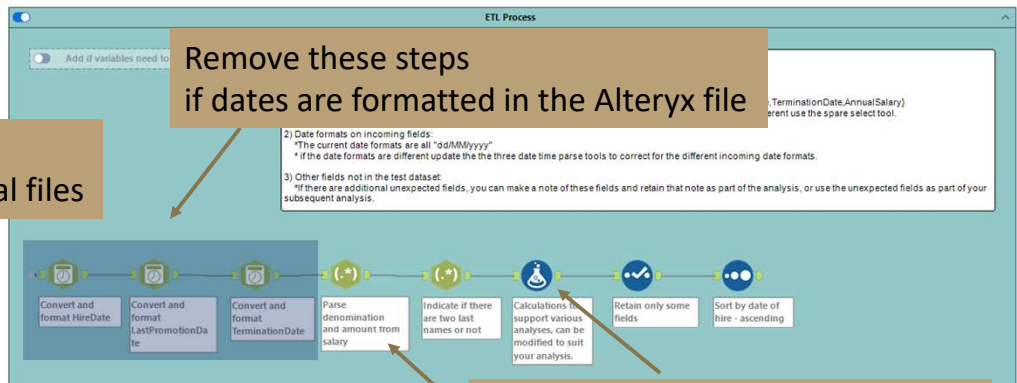
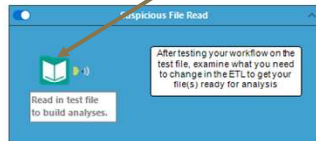
Convert dates to Alteryx format

ETL Challenge levels

1. Use single files, class_prompt (1), followup (2) and python (3)
2. Use a post ETL File (only if you get stuck)
3. Create a new file for the ETL workflow (se hints on logic).
4. Do the ETL in as few steps as possible.



Replace with
Alteryx file or individual files

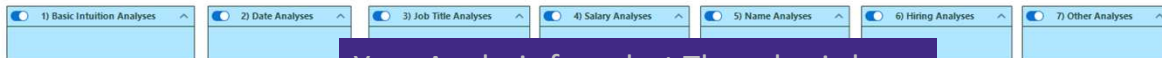


Remove these steps
if dates are formatted in the Alteryx file

Check that these steps are
working for the new file(s)

Use containers to categorize your tests. Consider tests in the form of:

- 1) Basic Intuition Analyses: Expected facts from the case.
- 2) Date Analyses: logical date sequences for events (i.e., some events cannot occur before/after other events)
- 3) Job Title and Salary Analyses: Logical company positions and structure at various ranks.
- 4) Salary Analyses: Logical salary hierarchy and variation.
- 5) Name Analyses: Consistency with expectations about case background.
- 6) Hiring Analyses: Logical company hiring/firing patterns.
- and 7) Other Analyses: Any other tests not defined above.



Your Analysis from last Thursday is here

ETL Logic

1. Use wildcard * read of csv files in folder, keep filename
2. Create source w/ regex
“submission_(\d)_\d+” (e.g., 1, 2, or 3)
3. Sort out the dates
4. Save file in Alteryx format

ETL Logic

1. Sort out the dates:

1. Replace / with – in all dates (replace() in formula tool)
2. Split to columns using –
3. Find max date part for each file (summarize tool)
4. Use logic to identify incoming date format $mm \leq 12$, $dd \leq 31$, $yyyy > 31$ (formula tool, **one expression)
5. Use logic to parse Alteryx Dates using dateparse() in formula

**Use multi-field formula to parse all three dates where possible

Thank you!

